

Final Exam

ORF 245 - Fundamentals of Statistics
Fall 2023

Date: Friday, December 22, 2023

Instructions:

- Please do not turn over this page until you are told to start the exam.
- Please write all of your solutions in the booklet, and make sure to write your name on both the cover of the booklet and the exam.
- The exam is 3 hours (i.e., from 9:00am - noon).
- Other than two 8.5x11 inch sheets of notes, this is a closed book exam. So, please do not use any other resources (book, notes, calculator, cell phone, etc.).
- The exam has two parts for a total of 90 points.
- Part I has 30 short answer questions worth one point each. You do **not** need to show any work (other than your answers) for the questions in Part I.
- Part II has six problems worth 10 points each. Please **show your work** for the questions in Part II.
- At the end of the exam, please put your exam inside of the booklet and return BOTH.
- No collaboration allowed. You should not discuss the exam questions with others until after the exam is over.
- Good luck!

Part I (1 point each, 30 points total)

You do **NOT** need to show any work (other than your answers) for this part.

1. True or False? For any two events, if $\mathbb{P}(B) > 0$ then $\mathbb{P}(A|B) > 0$.
2. True or False? For any two events, $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$.
3. If a student randomly guesses the answers to 10 True/False questions – where each possible answer is chosen with probability $1/2$ – what is the probability that the student will answer all 10 questions correctly?
4. With a standard 52-card deck, how many different 5-card poker hands are there?
5. True or False? If $X_1, \dots, X_n$ are Bernoulli random variables with success probability p , then $Y = \sum_{i=1}^n X_i$ has a Binomial(n, p) distribution.
6. True or False? For any random variable X the variance of X is always non-negative.
7. True or False? Let $f(x)$ denote the probability density function (pdf) of a continuous random variable X . Then $f(a) \leq f(b)$ whenever $a \leq b$.
8. True or False? If $F(x)$ and $F(y)$ denote the cdf's of random variables X and Y , respectively, then $F(x) + F(y)$ denotes the cdf of $X + Y$.
9. True or False? For any two random variables X and Y , $\text{Cov}(2X, Y) = 2 \text{Cov}(X, Y)$.
10. True or False? For any function $g(\cdot)$ and random variable X , $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$.
11. True or False? For any $a, b \in \mathbb{R}$ and random variable X , $\mathbb{E}[aX + b] = a \mathbb{E}[X] + b$.
12. True or False? According to the Central Limit Theorem, the sum of a large enough number of independent random variables has a Gaussian distribution.
13. If X is a Gaussian random variable with mean μ and variance σ^2 , what is $\mathbb{P}(X = \mu)$ (i.e., the probability that $X = \mu$)?
14. True or False? Point estimation is designed to yield a range of values within which we can be confident the true population parameter lies.
15. True or False? An estimator $\hat{\theta}$ of a parameter θ is unbiased if and only if $\mathbb{E}[(\hat{\theta} - \theta)^2] = 0$.

16. True or False? The maximum likelihood estimator for a parameter is always unbiased.
17. When building a confidence intervals for an unknown parameter – at a fixed confidence level (say 95%) – which of the following are random: the interval, the population mean, both, or neither?
18. True or False? For a fixed parameter θ and confidence level (α), there is only a single unique $1 - \alpha$ confidence interval for θ .
19. True or False? Two confidence intervals obtained from two independent samples can be non-overlapping.
20. True or False? When the population distribution is Gaussian and the population variance is known, doubling the sample size will halve the width of a two-sided confidence interval for the mean (of some fixed confidence level).
21. True or False? In hypothesis testing, the p-value is the probability that the null hypothesis H_0 is true.
22. True or False? In hypothesis testing, the p-value is computed under the assumption that the null hypothesis H_0 is true.
23. True or False? For a hypothesis test with significance level α , the probability of a Type I Error is equal to α .
24. True or False? For a hypothesis test with significance level α , the probability of a Type II Error is equal to $1 - \alpha$.
25. True or False? As its number of degrees of freedom increase, the t-distribution becomes more ‘spread out’ (i.e. has higher variance).
26. True or False? Let $t_{\alpha,\nu}$ be the critical values of a t-distribution with ν degrees of freedom, and z_α be critical values of the standard normal. Then $t_{5\%,10} > z_{5\%}$.
27. In linear regression, what is the range of possible values that the coefficient of determination, r^2 , can take on?
28. True or False? Under the standard assumptions for linear regression, the estimated coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased.

29. True or False? In linear regression, the *estimated sample* covariance between the residuals $\hat{\epsilon}_1, \dots, \hat{\epsilon}_n$ and $x_1, \dots, x_n$ is always zero.
30. Imagine that we run the regression $Y = \beta_0 + \beta_1 X + \epsilon$ using data $(x_1, y_1), \dots, (x_n, y_n)$ and obtain some estimates $\hat{\beta}_0$ and $\hat{\beta}_1$. If we define $Z = 2X$ and estimate the coefficients of $Y = \alpha_0 + \alpha_1 Z + \epsilon$ using data $(z_i, y_i) = (2x_i, y_i)$, what will be the estimate $\hat{\alpha}_1$ in terms of $\hat{\beta}_1$?

Part II (10 points each, 60 points total)

Please **show your work** for this part.

The following table gives approximate critical values of standard normal z_α and t-distributions $t_{\alpha,\nu}$ that might be useful for some of the computations:

α	z_α
0.100	1.282
0.050	1.645
0.025	1.960
0.010	2.326
0.005	2.576
0.001	3.090

(a) z -values

α				
ν	0.1	0.05	0.025	0.005
1	3.078	6.314	12.706	63.657
2	1.886	2.920	4.303	9.925
9	1.383	1.833	2.262	3.250
10	1.372	1.812	2.228	3.169
11	1.363	1.796	2.201	3.106
24	1.318	1.711	2.064	2.797

(b) $t_{\alpha,\nu}$ -values

1. (10 points)

Consider a bag with 1 red marble, 2 green marbles, and 3 blue marbles.

- (a) If you draw one marble out of the bag at random, what is the probability that the marble is blue?
- (b) If you draw two marbles out of the bag one at a time without replacement, what is the probability that both marbles are blue?
- (c) If you draw three marbles one at a time without replacement, what is the probability that none of the marbles are blue?
- (d) Calculate the expected number of draws without replacement needed to get a blue marble.

2. (10 points)

Let X be a random variable with pdf

$$f(x) = \frac{1}{\sqrt{\pi}} \exp(-(x + \pi)^2)$$

- (a) What is the mean $\mathbb{E}[X]$ and variance $\mathbb{V}[X]$ of X ?
- (b) Find $\mathbb{E}[X^2]$.
- (c) Let $Y = 2 - X/\pi$. Find the mean $\mathbb{E}[Y]$ and variance $\mathbb{V}[Y]$ of Y .
- (d) Find $\mathbb{P}[X > \pi]$ in terms of the cdf $\Phi(z)$ of the standard normal distribution.

3. (10 points)

Suppose that the discrete random variable X has a Poisson distribution with parameter $\lambda > 0$. That is, the pmf of X is given by

$$\mathbb{P}(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

and $\mathbb{P}(X = k) = 0$ otherwise. Recall that a Poisson random variable models the number of arrivals of some process in a given unit of time, and that the mean is given by λ .

- (a) If we observe 8 arrivals in one unit of time, find the likelihood function and the log likelihood.

Note: the log likelihood is defined as the logarithm of the likelihood function.

- (b) Find the maximum likelihood estimate of λ given that we observe 8 arrivals.
- (c) Now, suppose that we get n independent realizations $x_1, x_2, \dots, x_n$ of X . Find the likelihood and log likelihood functions.
- (d) Find the maximum likelihood estimate of λ given the n observations $x_1, x_2, \dots, x_n$.
- (e) Is this estimate biased or unbiased? Justify your answer.

4. (10 points)

Let $X_1, \dots, X_n$ be i.i.d. random variables according to a uniform distribution on the interval $[0, \theta]$, where θ is an unknown parameter. Their probability density function is given by:

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} & \text{if } x \in [0, \theta] \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Show that $\hat{\theta}_{\text{AVG}} = 2\bar{X}$ is an unbiased estimator of θ .
- (b) Compute $\text{Var}(\hat{\theta}_{\text{AVG}})$.
- (c) Let $X_{(n)} = \max_{i=1 \dots n} X_i$ denote the largest of all X_i 's. Its cumulative distribution function is given by:

$$F_{(n)}(x; \theta) = \begin{cases} 0 & \text{if } x < 0 \\ \left(\frac{x}{\theta}\right)^n & \text{if } x \in [0, \theta] \\ 1 & \text{if } x > \theta. \end{cases}$$

The maximum likelihood estimator for θ is given by $\hat{\theta}_{MLE} = X_{(n)}$.

Compute $\mathbb{E}[X_{(n)}] = \mathbb{E}[\hat{\theta}_{MLE}]$. Is this estimator biased?

- (d) Compute $\text{Var}(\hat{\theta}_{MLE})$.

5. (10 points)

Alexander Rd is one of the main roads connecting Princeton Campus with the Brunswick Pike and Princeton Junction Station. To evaluate the recent construction work, 81 commuters were asked to register their travel times from Nassau St to Princeton Junction Station, measured in minutes, the day before the road's closure and the day after its reopening. The result was:

- Before Construction:
 - Sample Average: $\hat{\mu}_b = 10.6$
 - Sample Standard Deviation: $\hat{s}_b = 0.9$
- After Construction:
 - Sample Average: $\hat{\mu}_a = 9.5$
 - Sample Standard Deviation: $\hat{s}_a = 1.2$

You may assume that (i) all reported travel times are outcomes of independent random variables; (ii) the samples of before and after construction are independent from one another.

- (a) What is the approximate distribution of difference between the sample average travel times before and after? I.e., Letting $X_1, \dots, X_{81}$ denote the travel times before, and $Y_1, \dots, Y_{81}$ the travel times after, what is the approximate distribution of $\bar{X} - \bar{Y}$?
- (b) Construct a lower one-sided 95% confidence interval for the mean travel time improvement $\mu_\Delta = \mu_b - \mu_a$.
- (c) The Municipality of Princeton anticipated that the construction would decrease the average travel time for commuters by more than 1/2 minute. Perform a statistical test about μ_Δ where you test

$$H_0 : \mu_\Delta = 1/2 \text{ min}$$

against

$$H_a : \mu_\Delta > 1/2 \text{ min}$$

at significance level 5%. Your conclusion about whether to reject H_0 , or not should be clearly stated and motivated.

Hint: You may use the table of critical values given earlier in the exam.

- (d) If the true value of μ_Δ is equal to 1, what is the probability of a Type II Error? You may leave your answer in terms of the standard normal CDF $\Phi(\cdot)$.

6. (10 points)

Consider a sample $(x_i, y_i)_{i=1}^{100}$ with the following quantities computed:

- $\sum_{i=1}^{100} x_i = 300$ $\sum_{i=1}^{100} y_i = 400$
- $\sum_{i=1}^{100} x_i^2 = 1300$ $\sum_{i=1}^{100} y_i^2 = 2000$
- $\sum_{i=1}^{100} x_i y_i = 1500$

- (a) Compute the estimate for $\hat{\beta}_1$ if the data follows $Y = \beta_0 + \beta_1 X + \epsilon$, and all the usual assumptions are satisfied.
- (b) Assume that our estimate for the variance of errors is given by $\hat{\sigma}_\epsilon^2 = 100$. Compute the standard error of $\hat{\beta}_1$, i.e. $\text{SE}(\hat{\beta}_1) = \sqrt{\text{Var}(\hat{\beta}_1)}$.
(You may assume that $x_1, \dots, x_n$ are treated as fixed)
- (c) At a 5% significance level, can we reject the null hypothesis $H_0 : \beta_1 = 0$? Assume that the alternative hypothesis is $H_A : \beta_1 \neq 0$.
- (d) Now (and in all subsequent parts), imagine that the data you observed has been mysteriously quadrupled by a bug in your code.

Namely, you now have $n = 400$ such that for every i between 1 and 100 we have $(x_i, y_i) = (x_{100+i}, y_{100+i}) = (x_{200+i}, y_{200+i}) = (x_{300+i}, y_{300+i})$. For example, $(x_1, y_1) = (x_{101}, y_{101}) = (x_{201}, y_{201}) = (x_{301}, y_{301})$.

Recompute the estimate $\hat{\beta}_1$ assuming that you use all 400 data points.

- (e) For simplicity, assume that our estimate for the variance of errors is still given by $\hat{\sigma}_\epsilon^2 = 100$. What is the new standard error of $\hat{\beta}_1$? (Again, using all 400 data points.)
Can you now reject the null hypothesis of $H_0 : \beta_1 = 0$ at a 5% significance level? Again, assume that the alternative hypothesis is $H_A : \beta_1 \neq 0$.
- (f) **Extra Credit: 2 points.** Explain why after quadrupling the data, we should never have the same value of $\hat{\sigma}_\epsilon^2$ as before. Namely, explain why the simplifying assumption made in part (e) is wrong.